Healthcare Governance & Compliance Report

Project: Hospital Operations & Revenue Risk Intelligence Platform
Phase: 6 — Monitoring, Drift Detection, and Governance

1. System Overview

This system consists of:

1. **Risk Model** – Classifies hospital visits as Low, Medium, or High operational/clinical risk.
2. **Claim Model** – Predicts insurance claim outcome as Paid, Pending, or Rejected.

The models are trained on historical hospital data and deployed via a FastAPI service for real-time predictions.

2. Assumptions

- Patient demographic and operational data are complete and accurate.
- Department values are limited to *ER, ICU,* and *OPD*.
- Financial features like *billed amount* and *payment days* reflect accurate billing information.
- Models are trained on data from [Hyderabad, Pune, Chennai, Bangalore, Mumbai, Delhi] and may not generalize to completely new locations.

3. Limitations

The system also has some limitations that we need to be aware of:

- Missing or invalid values in the data may make predictions less accurate.
- If a new or unknown category shows up (like a new department or insurance provider), it might cause errors.
- High-risk predictions might not be perfect, especially if some patient health information is missing.
- The models are trained on historical patterns, so sudden changes (like emergencies or unusual situations) might affect performance.

4. Retraining Strategy

To keep the models accurate, we retrain them regularly:

- **Frequency:** Retrain every 3 months, or sooner if we notice the model's predictions or features are drifting.
- **Trigger-based:** If the system detects significant changes in input data or prediction patterns, it will start a retraining process automatically.

- **Validation:** Every retraining is checked against historical and recent test data to make sure it works properly.
- **Versioning:** Each new model version is logged with its training date and the features used, so we can track changes over time.

5. Monitoring & Data Validation

We also have checks in place to make sure the system stays reliable:

- **Validation Checks:** The system checks numeric ranges, missing values, and categorical data before making predictions.
- **Feature Drift:** We monitor if numeric features change over time using KS-tests, which helps detect shifts from the training data.
- **Prediction Drift:** We keep track of the predicted class distributions to make sure they stay stable.
- **Audit Logs:** All predictions are logged in **prediction_audit_log.csv**. This helps with transparency and debugging.

6. Governance and Compliance

From my understanding, the system is designed to follow rules and regulations:

- Documentation is kept for all assumptions, limitations, and retraining steps.
- It supports compliance with regulations like HIPAA and internal audits by keeping records of predictions and outcomes.
- Audit logs can be checked if there is a prediction error or an operational incident.

7. Recommendations for Hospital Leadership

Based on how the system works, here are some recommendations:

- Periodically review model predictions with clinical staff to make sure they are consistent.
- Monitor drift reports and retrain models proactively if needed.
- Make sure all new departments or visit types are added correctly to the system.
- Keep sensitive patient data secure and only share anonymized or hashed information in logs.